Reinforcement Learning as a Risk-Budget Controller: Two Pre-Registered Walk-Forward Tests on Indian Equities

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Abstract

In our earlier study, a Proximal Policy Optimization (PPO) agent that chose the risk aversion and holding horizon of a mean-variance optimiser on seven Indian stocks lost to a passive index fund out of sample. Here the agent controls risk only. It chooses the exposure (50 to 100 per cent) and rebalance horizon (5 or 21 sessions) of an equal-risk-contribution portfolio, and cash earns the Reserve Bank of India's 91-day Treasury-bill rate. We ran two pre-registered walk-forward tests over 2019 to 2026, each with five seeds per fold combined in an ensemble and every strategy on one simulation engine. In the first, on a point-in-time Nifty 50 universe, the ensemble's Sharpe ratio beat volatility targeting by +0.320 (bootstrap 5th percentile +0.054), so its registered hypothesis was supported. Analyses made after that test found that the margin came from the 2020 crash and recovery, that the agent had learned to raise exposure after India VIX spikes, and that a two-parameter VIX threshold rule could not be distinguished from it. We then registered a second protocol that froze the whole pipeline, moved it to the point-in-time Nifty Next 50 and added the VIX rule as a second registered comparator under Holm's correction. Neither hypothesis held. The ensemble led volatility targeting by +0.195 ($p = 0.085$ against a threshold of 0.025), and its difference from the VIX rule was −0.028. The first result therefore did not replicate. On both universes the agent bought after fear spikes, a simple rule matched it, and its drawdowns were shallower than buy-and-hold; its edge over volatility targeting held on one universe only.

1 Introduction

Reinforcement learning (RL) has been applied to trading and portfolio choice for more than two decades [9, 15, 18, 22, 23, 40, 41]; Hambly et al. [11] and Pippas et al. [27] survey the field. Published backtests in this area are hard to compare and easy to overstate. Look-ahead in features or execution, universes chosen with hindsight, single training seeds, missing costs and an unreported number of tried configurations can each manufacture an apparent edge [4, 6, 14, 20].

Our own earlier study is a case in point [26]. A PPO agent chose the risk-aversion coefficient and holding horizon of a long-only mean-variance optimiser on seven large Indian stocks. Once the trading clock, data splits and costs were corrected, the agent lost to the NIFTYBEES index fund on its 2025–2026 forward period: a mean Sharpe ratio of −1.72 across seeds against −0.67 for the fund. A controller whose decisions depend on forecasting returns did not work.

This paper moves the agent from return forecasting to risk control. The agent never forecasts individual stocks. It decides how much of an equal-risk-contribution (ERC) portfolio to hold, between half and all of it, and how long to wait before deciding again. Three things recommend this design. Risk is more predictable than returns, which is the premise of volatility timing [10, 24]. A strong rule already answers the same question, volatility targeting [13], which gives

the agent a demanding and interpretable comparator. And a small discrete action space keeps the policy auditable.

We built the tests to be hard to fool. Both universes are rebuilt point in time from the exchange's own records: archived constituent files for the Nifty 50, and the index-change notices of NSE Indices, validated against every archived list we could find, for the Nifty Next 50. Cash earns the Treasury-bill rate of each date. Each protocol, including its hypotheses, was signed by all three authors and tagged in version control before any of its test years was scored, and every training run and analysis is recorded in an append-only trial ledger. Models are trained walk-forward, one fold per test year from 2019 to 2026, with five seeds per fold.

The first test, on the Nifty 50, supported its registered hypothesis, narrowly. The ensemble's Sharpe ratio exceeded that of volatility targeting by +0.320, with a bootstrap 5th percentile of +0.054. Analyses made after that test, which we label exploratory, showed that most of the margin was earned in 2020, that the agent had learned to hold little exposure when India VIX was low and to raise it after spikes, and that a VIX threshold rule designed after the test could not be distinguished from it. Findings made after looking cannot be claimed as confirmatory, so we registered a second protocol. It froze every design choice, applied the pipeline to the Nifty Next 50, the 50 stocks ranked below the Nifty 50, and registered the VIX rule as a comparator. On that universe the ensemble's lead over volatility targeting was +0.195 (95% interval [−0.088, +0.447]) and its difference from the VIX rule −0.028 ([−0.325, +0.213]). Neither hypothesis survived Holm's correction. Under the decision rule we registered in advance, the first result is reported as holding on one universe only.

Our contributions are:

- two pre-registered walk-forward evaluations of an RL risk-budget controller on point-in-time Indian equity universes, with code, data hashes, both protocols and a trial ledger;
- a registered positive result that a registered replication with the pipeline frozen did not reproduce, and the evidence for why: on both universes the agent learns contrarian volatility timing that a two-parameter rule matches; and
- a point-in-time Nifty Next 50 roster rebuilt from the exchange's index-change notices and validated against archived constituent lists, with every large price move checked against the exchange's daily files.

Section 2 reviews related work, Section 3 the data and Section 4 the method. Sections 5 to 7 report Study 1 on the Nifty 50: its registered evaluation, its results and the exploratory analyses that followed. Section 8 reports Study 2 on the Nifty Next 50, and Section 9 discusses both.

2 Related work

Reinforcement learning for trading and portfolios. Direct reinforcement for trading began with Moody et al. [23] and Moody and Saffell [22]; deep variants followed [9, 15, 41], as did ensembles of agents [40] and open libraries [18]. A related line keeps a conventional optimiser in the loop and lets learning steer it: RL over mean-variance optimisation [29, 35], and end-to-end risk budgeting with neural networks [38]. Sen et al. [34] compare mean-variance, hierarchical risk parity and RL on Indian equities. Our agent is a controller of this hybrid kind, restricted to the risk budget. Because it also chooses how long each decision lasts, its actions are temporally extended in the sense of semi-Markov options [37].

Volatility timing and targeting. Fleming et al. [10] show that timing volatility has economic value, and Moreira and Muir [24] report that scaling exposure by inverse recent variance raises Sharpe ratios across many factors. Harvey et al. [13] find that volatility targeting mainly

reduces tail risk, with Sharpe gains concentrated in risk assets. Out of sample the evidence is weaker: Cederburg et al. [7] find that volatility-managed portfolios do not systematically beat their unmanaged versions. Our primary comparator is volatility targeting with a 50 per cent exposure floor, matching the agent's own floor.

Risk parity and diversification. The ERC portfolio equalises each asset's contribution to portfolio risk [21]. We compute it with the convex formulation of Spinu [36] on the shrinkage covariance estimator of Ledoit and Wolf [16]. Naive diversification is a hard benchmark to beat [8], so equal weight is among our baselines.

Evaluation practice. Sharpe ratios are noisy and serially dependent [19]; we compare them with a paired block bootstrap [17, 28]. Selecting the best of many backtests inflates performance [4, 12]; the probabilistic and deflated Sharpe ratios correct for non-normality and for the number of trials [2, 3]. In RL, results vary widely across seeds [14], and point estimates without intervals mislead [1]. Pre-registration, standard in other empirical fields, commits the analysis before the data can shape it [25]; a registered replication on new data is the direct test of a finding that analysis after the fact has put in doubt.

3 Data

Universe. We rebuilt Nifty 50 membership from 76 archived captures of the National Stock Exchange's constituent files and index API, held by the Internet Archive, which contain 19 distinct member sets. Between captures, 78 additions and removals on 25 effective dates are dated from exchange notices or news reports, and renamed securities are linked by ISIN. The rebuilt roster matches 67 of the 68 snapshots taken on or after 1 January 2014; the one mismatch is a stale copy of the file from before a rename. The index has exactly 50 members on every session, or 51 while the Tata Motors DVR share was an additional security (April 2016 to September 2017).

Prices. Daily adjusted closes come from Yahoo Finance. Of the 87 symbols that were members between 2014 and the data freeze, 81 have usable series. The other 6 (CAIRN, HDFC, IDFC, JPASSOCIAT, RANBAXY, TATAMTRDVR) have none and are excluded while members; the largest is HDFC, a member until its merger in July 2023. A stock is eligible on a session when it is a member for the next session, has a price and has a complete 120-session return history; between 45 and 50 stocks are eligible on each session (median 49). We repaired 4 corporate-action errors, three demergers and one split-adjustment artifact, each from the exchange's own prices (Appendix C).

Benchmark, market series and cash. NIFTYBEES, an exchange-traded fund on the Nifty 50, is the passive benchmark, using its adjusted close as a total-return series; 3 sessions with vendor errors are repaired and 3 genuine tracking gaps are kept. The Nifty 50 index, India VIX and the rupee-dollar rate (lagged one session, because its close falls after the exchange's) feed the agent's state. Cash earns the cut-off yield of the RBI's weekly 91-day Treasury-bill auctions [31]: each yield applies from the session after its auction and accrues as $(1+y)^{d/365}-1$ over d calendar days. Over the test path the yield ranged from 2.93% to 7.05% (mean 5.33%).

Calendar and freeze. A session is any date with a Nifty 50 close on Yahoo Finance. This calendar skips a few exchange sessions, such as 1 January 2019, and a return then spans the skipped day; no return is lost. The data are frozen at the last session on or before 2026-11-06, and every raw and processed file is listed with its SHA-256 hash. Both studies use this calendar, this freeze and the same cash series.

The second universe. Study 2 uses the Nifty Next 50, the 50 stocks ranked 51 to 100 by free-float capitalisation; with the Nifty 50 it makes up the Nifty 100. Its archived constituent files are too sparse to rebuild membership on their own, so we rebuilt it from the exchange's index-change notices. We downloaded all 449 equity notices that NSE Indices published from October 2013 to the freeze and searched their text for the index. The changes come from 38 of them: the 26 semi-annual reviews from March 2014 to September 2026 (the March 2020 review was deferred to June because of COVID-19) and 12 ad hoc changes for mergers, demergers and replacements in the Nifty 50. That gives 308 dated changes on 37 dates. The roster matches all 37 archived readings from 2014 to 2026: 22 copies of the exchange's Nifty Next 50 file and 15 copies of its Nifty 100 file less the Study 1 roster. It holds exactly 50 members on every session and never overlaps the Nifty 50. Prices follow the Study 1 rules. Of the 152 members, 143 have usable series. The other 9 (ABIRLANUVO, CAIRN, GSKCONS, ICICIPRULI, INGVYSYABK, MINDTREE, PEL, RANBAXY, RELCAPITAL) are excluded while members, 2.8% of member-sessions in the test years; ICICI Prudential Life is among them only because the vendor's history now starts in July 2026, though the exchange's daily files show it trading throughout. A daily move beyond 25 per cent was accepted only if the exchange's own daily price file reproduced it and a cause was recorded; all 13 such moves passed (Appendix C). Vendor errors were repaired as in Study 1: one demerger from the exchange's pre-open price and 3 split-adjustment artifacts automatically. A post-test audit of these repairs against the exchange's files found one of the automatic ones wrong: on 4 June 2024 it replaced a genuine −19.2% fall in CONCOR, on the day of the general-election results, with +1.0%. CONCOR had left the index in March 2021 and was not eligible at any point the error could reach, so no result changes; the registered data were left as they were. The benchmark is JUNIORBEES, the exchange-traded fund on the index, and the Nifty Next 50 index replaces the Nifty 50 in the agent's index features. Between 45 and 50 stocks are eligible on each session (median 48). The two indices are closely related: their daily returns correlated 0.83 over 2014 to 2018.

4 Method

4.1 Allocator and engine

On each session t the allocator computes ERC weights w_t over the eligible stocks from the Ledoit-Wolf covariance Σ_t of the trailing 120 daily returns. Following Spinu [36], it solves

$$y^\star = \arg\min_{y>0} \ \tfrac{1}{2}\, y^\top \Sigma_t\, y - \tfrac{1}{N}\sum_{i=1}^{N} \log y_i, \qquad w_t = y^\star / \mathbf{1}^\top y^\star, \tag{1}$$

whose solution gives each stock the same contribution $w_i(\Sigma_t w)_i$ to portfolio variance. The weights do not depend on the agent's actions, so they are computed once for every session.

All strategies run on one engine. A decision at the close of session t sets the stock book to $e_t w_t$ of net asset value and holds $1 - e_t$ in cash; trades execute at that close and returns accrue from session $t + 1$. Each trade costs 10 basis points of the value traded. Holdings drift between decisions. A stock that leaves the index is sold at the close of its last member session. A decision is forced at each year-end close, so every test year is traded only by its own fold's models.

4.2 Agent

At each decision the agent picks an exposure $e \in \{0.5, 0.7, 0.85, 1.0\}$ and a horizon $h \in \{5, 21\}$ sessions until its next decision. Its reward for the interval is

$$R = 100 \left[\sum_{\tau=t+1}^{t+h} \left(\log(1 + r_\tau) - \log(1 + r_\tau^f)\right) - \kappa \max(0,\ D_{t+h} - D_t) \right], \tag{2}$$

where r_τ is the strategy's return net of costs, r_τ^f the cash return, D the magnitude of the strategy's drawdown from its running peak and $\kappa = 1$. The state has 13 features. Ten describe the market: the realised volatility of the universe's index (the Nifty 50 in Study 1, the Nifty Next 50 in Study 2) over 20 and 60 sessions and their ratio; index momentum over 21 and 63 sessions; the index drawdown from its 252-session high; the average pairwise correlation of eligible stocks over 60 sessions; the India VIX level and its five-session log change; and the 21-session change in the rupee-dollar rate. These are standardised with the mean and standard deviation of the fold's training years only and clipped at ± 5. Three describe the agent's own portfolio: its current exposure, drawdown and 20-session volatility.

We train with PPO [33] in Stable-Baselines3 [30], using a 64×64 network, learning rate 3×10^{-4}, 1,024 steps per update, minibatches of 64, ten epochs, discount 1, generalised advantage estimation with $\lambda = 0.95$ [32], clip range 0.2 and entropy coefficient 0.01. Training episodes are random 252-session windows within the training years, each starting in cash.

4.3 Walk-forward folds and the ensemble

Each test year Y from 2019 to 2026 has its own fold: training on 2014 to $Y - 3$ and validation on $Y - 2$ and $Y - 1$ (Appendix A). Each fold trains five seeds for 200,000 steps. Every 10,000 steps the current policy is run deterministically on each validation year, starting from cash, and the checkpoint with the best mean Sharpe ratio over the two years is kept. A run took 2.7 minutes on average on a laptop CPU. The ensemble keeps its own portfolio and state; at each of its decisions it queries all five seeds, takes the mean of their exposures and the median of their horizons.

4.4 Baselines

Every baseline uses the same engine, costs and execution, and starts in cash at the last close of 2018: NIFTYBEES bought and held; equal weight over eligible stocks, rebalanced at each month end; risk parity at a fixed exposure of 0.5, 0.7, 0.85 or 1.0, rebalanced every 21 sessions; the best fixed action, chosen per fold on validation; a random policy over the agent's eight actions (100 seeds); and volatility targeting. Volatility targeting sets $e_t = \text{clip}(\tau/\hat{\sigma}_t,\ 0.5,\ 1)$, where $\hat{\sigma}_t$ is the realised volatility of the unlevered ERC portfolio over 60 sessions; the target $\tau \in \{10, 12, 15\}$ per cent and horizon $h \in \{5, 21\}$ are chosen per fold on validation by the same criterion as the agent's checkpoints. The same rule without the floor is a secondary baseline.

5 Study 1: pre-registered evaluation on the Nifty 50

Registration. The protocol (`protocol.md`, mirrored by `config/protocol.yaml`, SHA-256 prefix `c97977a4a4bbb055`) was signed by all three authors and tagged `protocol-v1` on 7 November 2026, before any test year was scored. The test script refuses to run unless the tag exists, the configuration is unchanged since the tag, the working tree is clean and the cash series is the RBI one, and it writes its ledger entry before computing any test-year score. It was run once.

Primary hypothesis. H1: the PPO ensemble's Sharpe ratio exceeds that of floored volatility targeting over the combined test path. Sharpe ratios are annualised: $\sqrt{252}$ times the mean daily excess return over its standard deviation. We resample the paired daily returns of the two strategies with a circular block bootstrap [28] in blocks of 20 sessions, 10,000 times with a fixed seed. H1 is supported if the 5th percentile of the bootstrapped Sharpe difference is above zero.

Secondary outcomes and sensitivities. The secondary outcomes are maximum drawdown and Calmar ratio against equal weight and NIFTYBEES. Four sensitivities were registered:

executing every trade one session late, doubling costs to 20 basis points, the unfloored volatility rule, and a 30 per cent loss on any stock that stopped trading while a member (none did on the test path, so it does not apply). Every other comparison is exploratory.

Multiple testing. We report the deflated Sharpe ratio (DSR) of Bailey and López de Prado [3]: the probability that the true Sharpe ratio exceeds the expected maximum of N unskilled trials, with the null variance of the Sharpe ratio set to $1/(T-1)$ and the non-normality correction of Bailey and López de Prado [2]. N is taken from the ledger, once for this study alone and once including the trials of our earlier work (Appendix B).

Correctness checks. Before any training, the test suite established that perturbing every series after a date leaves all features, weights, standardisation statistics, decisions and net asset values before it bit-identical; that an oracle policy shown the next interval's return earns an absurd Sharpe ratio, and loses its edge when lagged one interval, so returns are credited after decisions; that costs match hand-computed trades; that the allocator returns inverse-volatility weights for a diagonal covariance and equal risk contributions to within 10^{-4}; and that the statistics reproduce published examples. Development runs used only 2014 to 2018.

6 Study 1: results

6.1 The primary test

Table 1 reports the combined path. The ensemble's Sharpe ratio is 0.808, against 0.488 for floored volatility targeting. The bootstrapped difference is +0.320 with a 5th percentile of +0.054 (95% interval [+0.000, +0.620]; 2.5% of draws at or below zero), so H1 is supported under the registered rule. The margin is thin: the lower end of the two-sided interval is zero.

The ensemble earned 14.9% a year with 11.4% volatility, while holding on average 0.66 of the portfolio in stocks against 0.87 for volatility targeting. Its maximum drawdown was −20.7%, against −29.9% for volatility targeting, −38.3% for equal weight and −36.3% for NIFTYBEES; only risk parity held at a constant 0.5 exposure fell less (−17.4%). On the registered secondary outcomes, the ensemble's maximum drawdown was shallower than equal weight's by +17.6 percentage points (95% interval [+4.1, +24.9]) and than NIFTYBEES's by +15.7 ([+3.2, +27.8]), and its Calmar ratio was higher by +0.292 [+0.050, +0.808] and +0.373 [+0.098, +1.151] respectively. Figures 1 and 2 show the paths.

Table 2 sets the ensemble against every baseline. Apart from H1, these comparisons are exploratory. Apart from the two volatility-targeting rules, every interval includes zero: against equal weight the difference is +0.147 [−0.078, +0.345], against NIFTYBEES +0.305 [−0.041, +0.608], and against risk parity at 0.7 exposure, the fixed exposure nearest the ensemble's average, +0.190 [−0.015, +0.379]. Volatility targeting's Sharpe ratio, 0.488, is below that of risk parity at every fixed exposure (0.624 to 0.611).

6.2 Year by year

The ensemble is ahead of volatility targeting in 5 of the 8 test years (Figure 3, Table 3), but no single year's interval excludes zero. The largest gap is 2020, at +1.175 [−0.117, +1.847]. That fold's checkpoints all had negative validation Sharpe ratios (−0.15 to −0.07) on 2018–2019, yet its models produced the ensemble's best test year. In 2019 every risk-parity strategy and equal weight trailed NIFTYBEES, which weights stocks by market capitalisation.

Table 1: The combined test path, 2018-12-31 to 2026-11-06 (1,936 sessions). Costs are the total paid over the path as a share of capital; exposure is the mean invested share at decisions.

Strategy	Sharpe	CAGR	Vol.	Max DD	Calmar	Costs	Exposure
PPO ensemble	0.808	14.9%	11.4%	−20.7%	0.72	2.7%	0.66
Vol targeting (floor 0.5)	0.488	11.4%	12.9%	−29.9%	0.38	2.7%	0.87
Vol targeting (no floor)	0.483	11.5%	13.5%	−29.9%	0.39	2.8%	0.93
Best fixed action	0.547	12.2%	12.9%	−34.9%	0.35	2.0%	0.65
Risk parity, $e = 0.50$	0.624	10.4%	7.8%	−17.4%	0.60	0.8%	0.50
Risk parity, $e = 0.70$	0.619	12.2%	11.0%	−24.3%	0.50	1.2%	0.70
Risk parity, $e = 0.85$	0.615	13.5%	13.4%	−29.7%	0.45	1.6%	0.85
Risk parity, $e = 1.00$	0.611	14.7%	15.8%	−35.0%	0.42	2.0%	1.00
Equal weight (monthly)	0.662	16.4%	17.2%	−38.3%	0.43	1.2%	1.00
NIFTYBEES buy-and-hold	0.503	12.7%	15.5%	−36.3%	0.35	0.1%	1.00
Random policy (median of 100)	0.567						

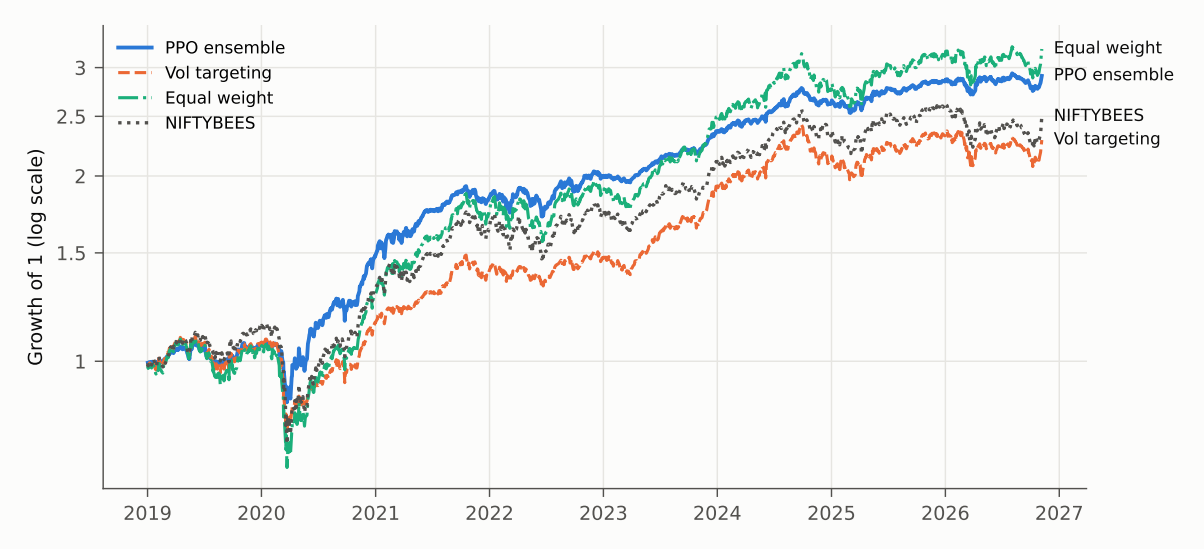


Figure 1: Growth of one rupee over the test path, log scale.

6.3 Robustness

All 5 seeds beat volatility targeting on their own, with Sharpe ratios from 0.611 to 0.876 (Table 4, Figure 4). Executing every trade one session late leaves the ensemble's Sharpe ratio at 0.808 (volatility targeting: 0.532); an edge that relied on same-session information would shrink under the delay. With costs doubled to 20 basis points, the ensemble's Sharpe ratio is 0.791 and volatility targeting's 0.470. The unfloored volatility rule scores 0.483. One hundred random policies over the same actions have a median Sharpe ratio of 0.567 and a 5th to 95th percentile range of 0.410 to 0.759; the ensemble lies above that range.

The ensemble's daily Sharpe ratio is 0.0509, with skewness +0.05 and kurtosis 20.4. With the single trial this study had run at the time of the test, its DSR is 0.987. Counting all 83 trials in the ledger, including those of our earlier work, it is 0.412. After the post-hoc rule of Section 7.3, which we count as a trial, the figures are 0.957 (2 trials) and 0.410 (84 trials); Table 14 gives them at the project's final count.

6.4 Drawdown episodes

The protocol fixed two windows: NIFTYBEES's deepest fall whose peak lies in 2020, and the same for 2025 (Table 5). From 2020-01-17 to 2020-03-23, the ensemble lost −20.0%, against

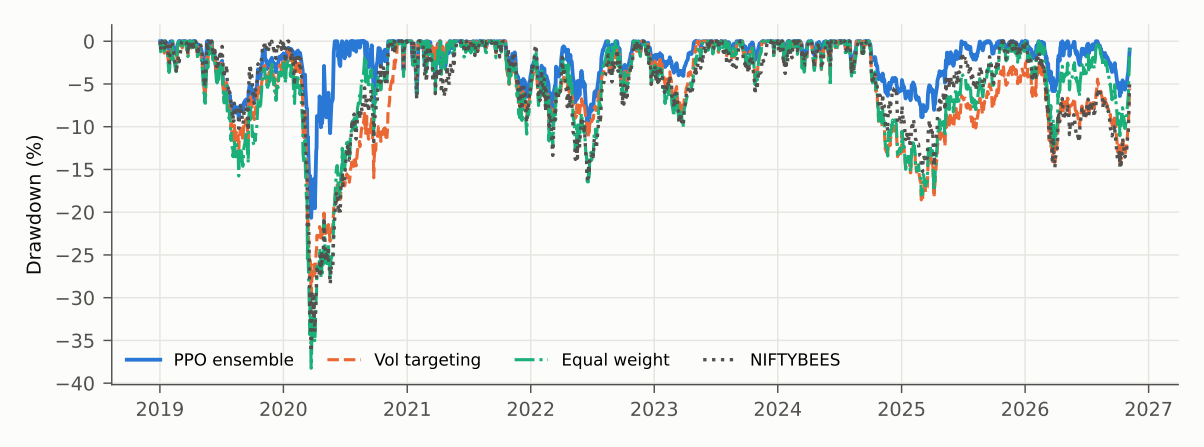


Figure 2: Drawdown from the running peak.

Table 2: Sharpe difference, PPO ensemble minus each baseline, over the whole path. Only the first row is the pre-registered test; the last is the post-hoc rule of Section 7.3.

Baseline	PPO − baseline	95% interval	Draws
Vol targeting (floor 0.5) (H1)	+0.320	[+0.000, +0.620]	10,000
Vol targeting (no floor)	+0.326	[+0.034, +0.636]	2,000
Best fixed action	+0.261	[−0.149, +0.550]	2,000
Equal weight (monthly)	+0.147	[−0.078, +0.345]	2,000
NIFTYBEES buy-and-hold	+0.305	[−0.041, +0.608]	2,000
Risk parity, $e = 0.50$	+0.185	[−0.016, +0.374]	2,000
Risk parity, $e = 0.70$	+0.190	[−0.015, +0.379]	2,000
Risk parity, $e = 0.85$	+0.193	[−0.016, +0.385]	2,000
Risk parity, $e = 1.00$	+0.197	[−0.015, +0.391]	2,000
VIX threshold rule (post hoc)	+0.151	[−0.220, +0.417]	10,000

−29.1% for volatility targeting, −36.7% for equal weight and −36.3% for NIFTYBEES. The 2025 window turned out to be a small dip (−2.1% for NIFTYBEES), because the larger 2024–25 correction peaked in September 2024. Over that correction, from 2024-09-27 to 2025-03-04, the ensemble lost −8.9%, against −18.6% for volatility targeting and −15.2% for NIFTYBEES.

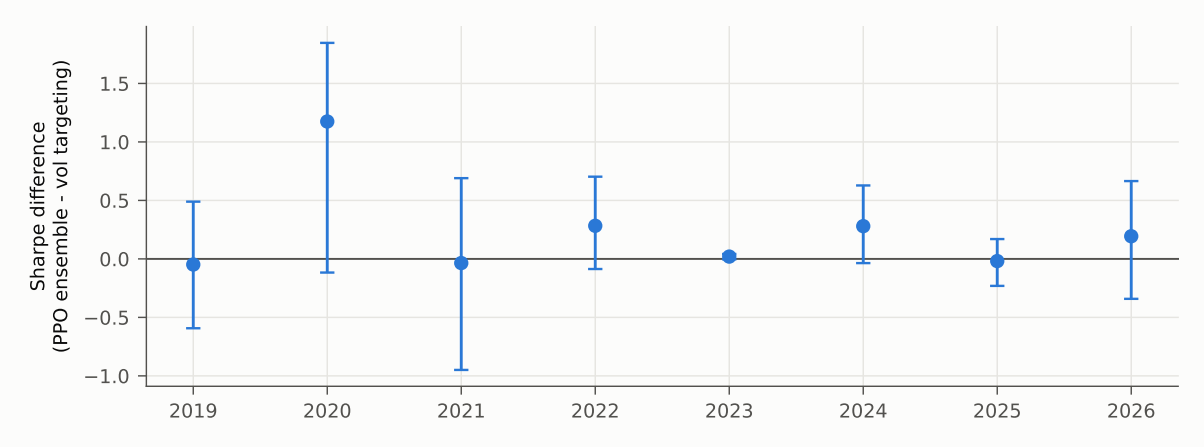


Figure 3: Sharpe difference by test year, PPO ensemble minus volatility targeting, with 95% bootstrap intervals.

Table 3: Sharpe ratio by test year. †Previously seen: our earlier work used these years.

Year	PPO	Vol tgt	Best fixed	Equal wt	NIFTYBEES	PPO − VT	95% interval
2019	0.077	0.125	0.139	0.037	0.618	−0.048	[−0.593, +0.489]
2020	1.491	0.317	0.707	0.699	0.533	+1.175	[−0.117, +1.847]
2021	1.630	1.666	1.780	1.925	1.473	−0.036	[−0.949, +0.690]
2022†	0.308	0.025	0.145	0.247	0.106	+0.283	[−0.086, +0.703]
2023†	2.400	2.380	2.376	2.232	1.452	+0.020	[−0.004, +0.042]
2024†	0.539	0.259	0.287	0.328	0.344	+0.280	[−0.036, +0.628]
2025†	0.483	0.502	0.532	0.693	0.544	−0.019	[−0.231, +0.170]
2026†	−0.379	−0.573	−0.537	−0.136	−0.624	+0.194	[−0.341, +0.665]

Table 4: Pre-registered robustness checks: single seeds, execution one session late and costs doubled.

Variant	Sharpe	CAGR	Max DD	Costs
PPO seed 0	0.809	14.9%	−18.4%	3.1%
PPO seed 1	0.681	13.0%	−18.1%	3.2%
PPO seed 2	0.876	15.9%	−18.1%	3.0%
PPO seed 3	0.810	15.0%	−17.5%	3.4%
PPO seed 4	0.611	13.1%	−30.9%	2.6%
PPO ensemble, one session late	0.808	15.0%	−22.8%	2.7%
Vol targeting, one session late	0.532	12.0%	−28.6%	2.8%
PPO ensemble, 20 bp costs	0.791	14.7%	−20.9%	5.2%
Vol targeting, 20 bp costs	0.470	11.1%	−30.1%	5.4%
Equal weight, 20 bp costs	0.656	16.3%	−38.3%	2.4%

Table 5: Return from peak to trough over NIFTYBEES drawdown windows. The protocol fixed the deepest windows whose peaks fall in 2020 and 2025; the other rows are the deepest on the whole path not already listed (exploratory).

Window (peak to trough)	Chosen as	PPO	Vol tgt	Equal wt	NIFTYBEES
2020-01-17 to 2020-03-23	Protocol, peak in 2020	−20.0%	−29.1%	−36.7%	−36.3%
2025-10-29 to 2025-11-06	Protocol, peak in 2025	−1.1%	−2.2%	−2.3%	−2.1%
2021-10-18 to 2022-06-17	Exploratory, deepest no. 2	−10.8%	−11.2%	−16.4%	−16.1%
2024-09-27 to 2025-03-04	Exploratory, deepest no. 3	−8.9%	−18.6%	−17.9%	−15.2%

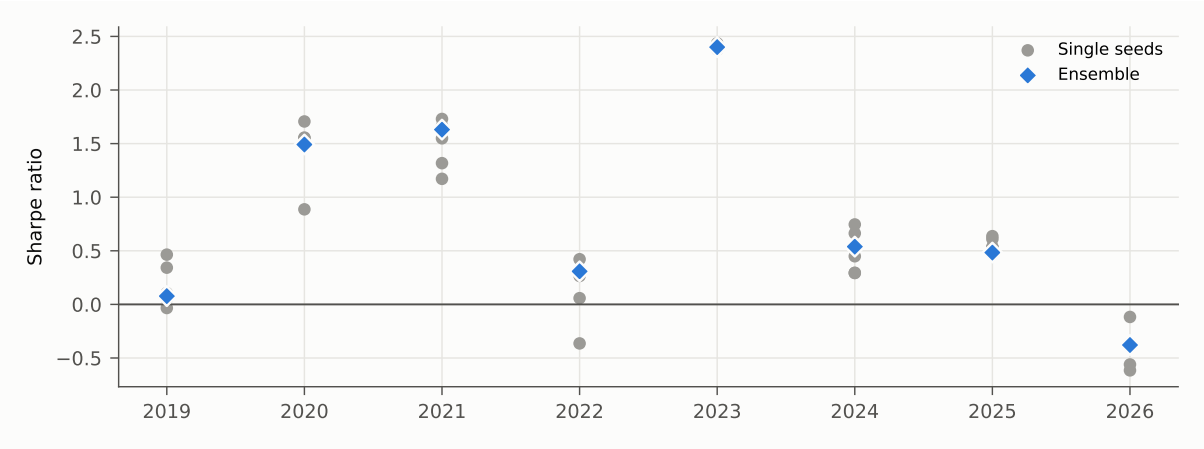


Figure 4: Sharpe ratio by test year for each seed and for the ensemble.

7 Study 1: exploratory analyses

Everything in this section was decided after the Study 1 test run. Each analysis is recorded in the ledger, and none can change the verdict on H1. Together they motivated Study 2.

7.1 What the agent learned

Across its 118 decisions, the ensemble's chosen exposure correlates +0.62 with India VIX at the decision and −0.29 with the Nifty 50 drawdown from its 252-session high (Figure 5, Table 6). With VIX below 15 its mean exposure was 0.54, at the floor in 71% of decisions; with VIX at 25 or above it was 0.87. The agent learned to hold little in calm markets and more after fear spikes. Volatility targeting does the reverse by construction, and its 60-session volatility estimate keeps exposure at the floor while a spike remains in the window.

The 2020 crash shows the difference. The ensemble's decisions were 0.50 on 29 January; 0.60 on 28 February; 1.00 on 31 March. It held half to 60 per cent of the portfolio through the fall and moved to full exposure eight days after the index bottomed on 2020-03-23, before most of the recovery.

7.2 How much rests on 2020

Without 2020, the ensemble's lead over volatility targeting falls to +0.101, with a 5th percentile of −0.149 (Table 7). On the previously seen years 2022 to 2026 it is +0.045 (5th percentile −0.260). On 2019 to 2021, the years our earlier work never used, it is +0.607 (5th percentile +0.114), but that window contains 2020. Outside 2020, monthly equal weight has a higher Sharpe ratio than the ensemble (0.691 against 0.649).

7.3 A VIX threshold rule

To ask whether the agent does more than buy after VIX spikes, we designed one rule after the test, committed it to the repository before running it, recorded it in the ledger before it touched any test year, and count it as a trial. The rule holds full exposure when India VIX at the decision close is at or above a threshold and the 0.5 floor otherwise; the threshold (15, 20 or 25) and horizon (5 or 21 sessions) are chosen per fold on validation, like the other tuned baselines.

The rule's Sharpe ratio is 0.658 (Table 8). The ensemble's lead over it is +0.151, with a 95% interval of [−0.220, +0.417], so the two cannot be told apart; the rule's own lead over volatility targeting, +0.169 [−0.122, +0.535], is not significant either. Outside 2020 the rule is ahead of the ensemble (0.742 against 0.649); in 2020 the ensemble is ahead (1.491 against 0.841). Between January and April 2020 the two made their 3 decisions on the same dates and differed once: on 28 February 2020, with India VIX at 23.2, the rule moved to exposure 1.00 and the ensemble to 0.60. The rule's maximum drawdown is −30.6%. Their exposures correlate +0.37 session by session, so the ensemble is not simply this rule; but most of its measured edge over the rule comes from 2020, where the two policies differ by a single decision. Because the rule was designed after the test, Study 2 registered it in advance as a comparator.

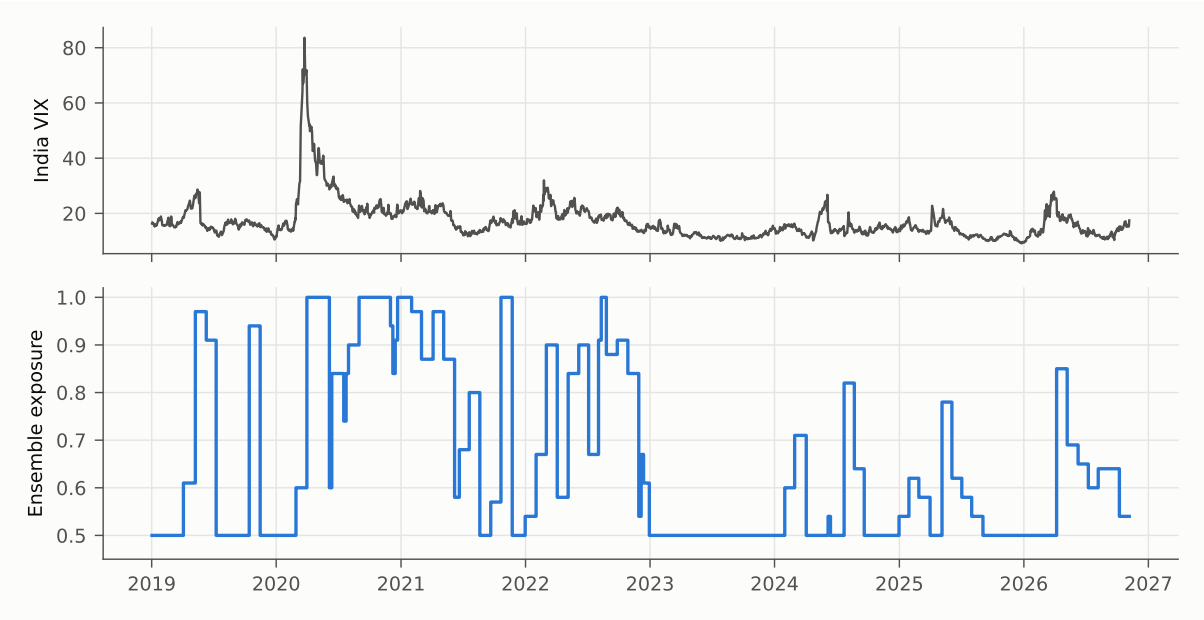


Figure 5: India VIX (top) and the exposure chosen by the PPO ensemble (bottom), held from each decision to the next.

Table 6: The ensemble's decisions grouped by India VIX at the decision (exploratory).

India VIX at the decision	Decisions	Mean exposure	At the floor
below 15	51	0.54	71%
15 to 20	38	0.67	32%
20 to 25	20	0.86	5%
25 and above	9	0.87	0%

Table 7: Sharpe differences by period, PPO ensemble minus each baseline, with the H1 bootstrap settings (exploratory). Seen: previously seen years.

Period	Compared with	PPO	Baseline	Difference	5th pct.	95% interval
2019–2021 (unseen)	Vol targeting (floor 0.5)	1.162	0.555	+0.607	+0.114	[+0.017, +1.055]
2019–2021 (unseen)	Risk parity, $e = 0.70$	1.162	0.825	+0.337	−0.001	[−0.055, +0.633]
2019–2021 (unseen)	Equal weight (monthly)	1.162	0.799	+0.363	−0.008	[−0.079, +0.686]
2019–2021 (unseen)	NIFTYBEES buy-and-hold	1.162	0.757	+0.405	−0.154	[−0.258, +0.891]
2022–2026 (seen)	Vol targeting (floor 0.5)	0.487	0.443	+0.045	−0.260	[−0.316, +0.413]
2022–2026 (seen)	Risk parity, $e = 0.70$	0.487	0.451	+0.036	−0.139	[−0.171, +0.240]
2022–2026 (seen)	Equal weight (monthly)	0.487	0.561	−0.074	−0.253	[−0.287, +0.137]
2022–2026 (seen)	NIFTYBEES buy-and-hold	0.487	0.285	+0.202	−0.084	[−0.146, +0.545]
All but 2020	Vol targeting (floor 0.5)	0.649	0.548	+0.101	−0.149	[−0.194, +0.400]
All but 2020	Risk parity, $e = 0.70$	0.649	0.611	+0.038	−0.114	[−0.143, +0.221]
All but 2020	Equal weight (monthly)	0.649	0.691	−0.042	−0.205	[−0.240, +0.158]
All but 2020	NIFTYBEES buy-and-hold	0.649	0.526	+0.123	−0.125	[−0.173, +0.410]

Table 8: The post-hoc VIX threshold rule against the PPO ensemble and volatility targeting, by period. "First" and "second" are the two strategies of each comparison, in order. Seen: previously seen years.

Period	Comparison	First	Second	Difference	5th pct.	95% interval
2019–2026	PPO − VIX rule	0.808	0.658	+0.151	−0.176	[−0.220, +0.417]
2019–2026	VIX rule − vol targeting	0.658	0.488	+0.169	−0.070	[−0.122, +0.535]
2019–2021 (unseen)	PPO − VIX rule	1.162	0.748	+0.414	−0.123	[−0.182, +0.774]
2019–2021 (unseen)	VIX rule − vol targeting	0.748	0.555	+0.193	−0.170	[−0.250, +0.746]
2022–2026 (seen)	PPO − VIX rule	0.487	0.646	−0.159	−0.436	[−0.489, +0.176]
2022–2026 (seen)	VIX rule − vol targeting	0.646	0.443	+0.203	−0.069	[−0.120, +0.531]
All but 2020	PPO − VIX rule	0.649	0.742	−0.092	−0.317	[−0.360, +0.183]
All but 2020	VIX rule − vol targeting	0.742	0.548	+0.193	−0.018	[−0.060, +0.451]

8 Study 2: a registered replication on the Nifty Next 50

8.1 Protocol

The analyses of Section 7 were made after the test, so they cannot be claimed as confirmatory. We therefore registered a second protocol (`protocol_v2.md`, mirrored by `config/protocol_v2.yaml`, whose SHA-256 begins `217c789dd2a95d9d`). All three authors signed it, and it was tagged `protocol-v2` on 8 November 2026, before any Nifty Next 50 test year was scored. It freezes every element of Study 1: the allocator, costs, execution, actions, reward, state, PPO settings, seeds, checkpoint selection, ensemble rule, folds, baselines, statistics and data freeze. A unit test fails if the configuration differs from Study 1's anywhere other than in the registered changes. Those are the universe and its index and benchmark (Section 3), and two additions.

First, the VIX threshold rule of Section 7.3 is a registered baseline, with the grid committed before the rule was first run. It holds full exposure when India VIX at the decision close is at or above a threshold and 0.5 otherwise; the threshold (15, 20 or 25) and horizon (5 or 21 sessions) are chosen per fold on validation, like every tuned baseline. Second, there are two primary hypotheses:

- H1: the ensemble's Sharpe ratio exceeds that of floored volatility targeting on the combined test path (a replication of Study 1's H1);
- H2: the ensemble's Sharpe ratio exceeds that of the VIX threshold rule.

Each is tested with Study 1's paired block bootstrap, and its p-value is the share of draws at or below zero. Holm's procedure holds the family-wise error rate at 0.05: the smaller p must be at most 0.025 and, if it is, the larger at most 0.05. The protocol also fixed what the paper would say for each outcome (Table 9). Every test year is labelled previously seen, because Study 1 examined the same market path. Before registration we ran one development training on fold 2019's years (2014 to 2018) and a dry run of the test code on 2017 and 2018; nothing was scored on 2019 onward.

Table 9: The decision rule registered with Study 2, before its test.

H1	H2	What the paper reports
Supported	Supported	The edge over volatility targeting replicates on new stocks, and PPO beats the simplest rule describing its behaviour.
Supported	Not supported	The edge replicates, but there is no evidence that PPO does more than a VIX threshold rule.
Not supported	Supported	Reported as is; Study 1's H1 stands as a result on one universe.
Not supported	Not supported	The Study 1 result does not replicate. The abstract says so, and Study 1's H1 is reported as holding on one universe only.

8.2 Results

Table 10: Study 2's primary hypotheses, tested once on the combined path with Holm's procedure at a family-wise α of 0.05.

	Comparison	Difference	95% interval	p	Holm threshold	Decision
H1	PPO ensemble − Vol targeting (floor 0.5)	+0.195	[−0.088, +0.447]	0.085	0.025	not supported
H2	PPO ensemble − VIX threshold rule	−0.028	[−0.325, +0.213]	0.613	0.050	not supported

Neither hypothesis is supported (Table 10). The ensemble's Sharpe ratio exceeds that of volatility targeting by +0.195 (95% interval [−0.088, +0.447]), with $p = 0.085$ against a Holm threshold of 0.025; it would not have passed Study 1's single-hypothesis rule either. Against the VIX threshold rule the difference is −0.028 ([−0.325, +0.213], $p = 0.613$). The last row of Table 9 applies.

Table 11: Study 2's combined test path, 2018-12-31 to 2026-11-06 (1,936 sessions). Columns as in Table 1.

Strategy	Sharpe	CAGR	Vol.	Max DD	Calmar	Costs	Exposure
PPO ensemble	0.639	13.8%	13.3%	−27.5%	0.50	2.8%	0.73
Vol targeting (floor 0.5)	0.444	11.0%	13.6%	−33.8%	0.33	2.5%	0.78
Vol targeting (no floor)	0.413	10.7%	14.2%	−33.8%	0.32	2.5%	0.87
VIX threshold rule (registered)	0.667	13.6%	12.3%	−30.0%	0.45	3.0%	0.61
Best fixed action	0.507	10.3%	9.7%	−17.2%	0.60	1.4%	0.53
Risk parity, $e = 0.50$	0.602	10.5%	8.3%	−17.2%	0.61	1.1%	0.50
Risk parity, $e = 0.70$	0.600	12.4%	11.7%	−24.0%	0.52	1.6%	0.70
Risk parity, $e = 0.85$	0.598	13.7%	14.3%	−28.9%	0.47	2.1%	0.85
Risk parity, $e = 1.00$	0.596	15.0%	16.9%	−33.8%	0.44	2.6%	1.00
Equal weight (monthly)	0.552	14.9%	18.7%	−37.0%	0.40	1.8%	1.00
JUNIORBEES buy-and-hold	0.514	13.6%	17.4%	−34.5%	0.39	0.1%	1.00
Random policy (median of 100)	0.538						

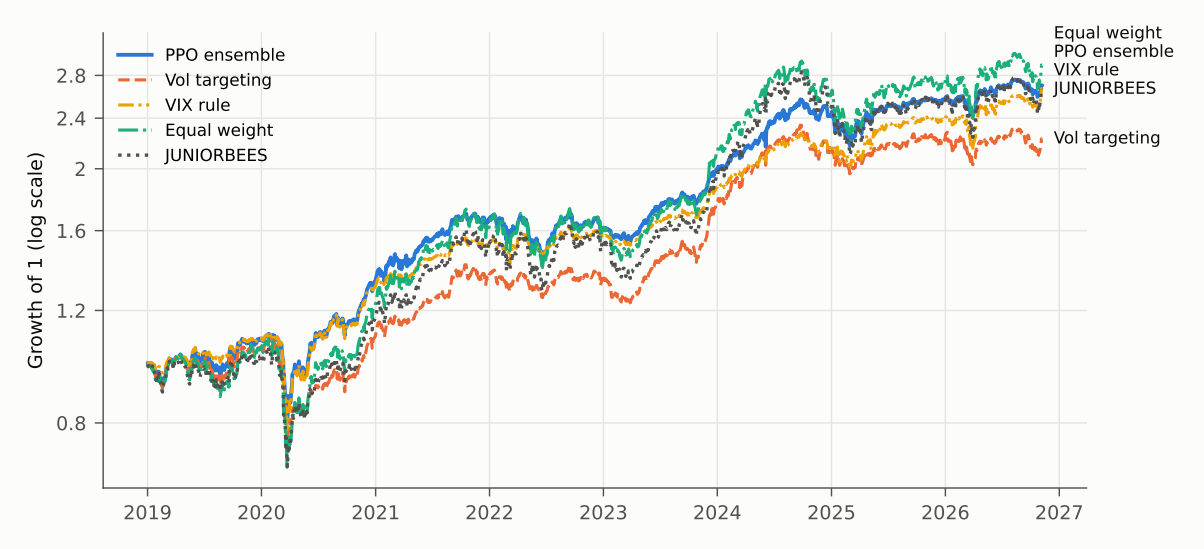


Figure 6: Study 2: growth of one rupee over the test path, log scale.

Table 11 and Figure 6 report the combined path. The ensemble's Sharpe ratio, 0.639, is below the VIX rule's 0.667 and only slightly above risk parity at a fixed exposure (0.596 to 0.602). It lies inside the 5th to 95th percentile range of the random policies (0.383 to 0.688, median 0.538). The two volatility-targeting rules are again the weakest risk-parity strategies (0.444 with the floor and 0.413 without; average exposure 0.78 and 0.87). The ensemble's maximum drawdown, −27.5%, is shallower than volatility targeting's (−33.8%), the VIX rule's (−30.0%) and JUNIORBEES's (−34.5%). On the registered secondary outcomes it was shallower than equal weight's by +9.5 percentage points (95% interval [+4.3, +18.6]) and than JUNIORBEES's by +7.0 ([+3.9, +20.0]). The Calmar differences, +0.100 [−0.043, +0.539] and +0.109 [−0.022, +0.671], are not significant.

By year (Table 12, Figure 7), the ensemble is ahead of volatility targeting in 8 of the 8 years, with intervals excluding zero in 2019 and 2026; the largest gap is again 2020 (+0.758 [−0.068,

Table 12: Study 2: Sharpe ratio by test year, with each year's difference and 95% bootstrap interval against volatility targeting and the VIX rule. Every year was previously seen in Study 1.

Year	PPO	Vol tgt	VIX rule	Equal wt	JUNIORBEES	PPO − VT [95%]	PPO − VIX rule [95%]
2019	0.229	−0.024	0.324	0.004	−0.260	+0.253 [+0.031, +0.529]	−0.095 [−0.560, +0.346]
2020	0.906	0.148	0.755	0.577	0.577	+0.758 [−0.068, +1.314]	+0.151 [−0.042, +0.295]
2021	1.840	1.834	1.860	1.766	1.632	+0.007 [−0.747, +0.720]	−0.020 [−0.476, +0.448]
2022	−0.498	−0.576	−0.181	−0.266	−0.132	+0.078 [−0.330, +0.572]	−0.317 [−0.971, +0.320]
2023	2.172	2.032	2.058	1.767	1.503	+0.140 [−0.399, +0.632]	+0.114 [−0.408, +0.605]
2024	1.171	0.979	1.040	0.878	1.204	+0.193 [−0.408, +0.804]	+0.132 [−0.290, +0.599]
2025	0.090	0.002	0.487	0.048	−0.108	+0.088 [−0.353, +0.546]	−0.397 [−0.904, +0.213]
2026	0.102	−0.423	0.602	0.189	0.057	+0.526 [+0.009, +0.991]	−0.500 [−1.257, +0.121]

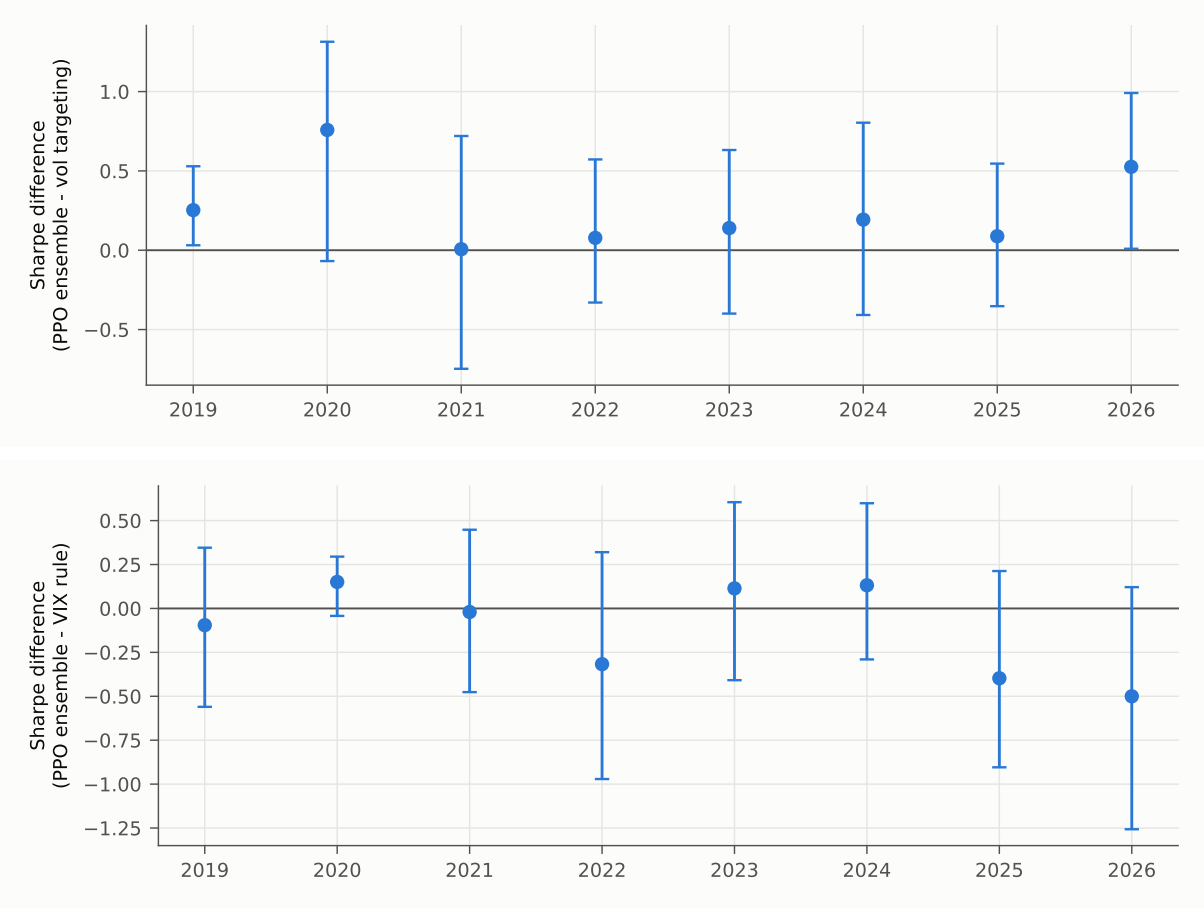


Figure 7: Study 2: Sharpe difference by test year, PPO ensemble minus volatility targeting (top) and minus the VIX rule (bottom), with 95% bootstrap intervals.

+1.314]). It is ahead of the VIX rule in 3 of the 8 years, 2020 among them (+0.151). As in Study 1, fold 2020's checkpoints all had negative validation Sharpe ratios (−0.15 to −0.05).

Table 13: Study 2: pre-registered robustness checks.

Variant	Sharpe	CAGR	Max DD	Costs
PPO seed 0	0.630	14.0%	−30.0%	3.3%
PPO seed 1	0.658	14.0%	−30.1%	3.7%
PPO seed 2	0.770	14.6%	−17.2%	3.9%
PPO seed 3	0.559	13.1%	−30.0%	3.3%
PPO seed 4	0.576	13.5%	−30.0%	3.6%
PPO ensemble, one session late	0.626	13.6%	−27.2%	2.8%
Vol targeting, one session late	0.461	11.2%	−33.6%	2.5%
VIX rule, one session late	0.662	13.5%	−29.7%	3.0%
PPO ensemble, 20 bp costs	0.622	13.6%	−27.5%	5.6%
Vol targeting, 20 bp costs	0.427	10.7%	−33.8%	4.9%
VIX rule, 20 bp costs	0.649	13.4%	−30.1%	6.0%
Equal weight, 20 bp costs	0.545	14.7%	−37.0%	3.6%

All 5 seeds beat volatility targeting, but only 1 beat the VIX rule; seed Sharpe ratios range from 0.559 to 0.770 (Table 13). With every trade one session late the ensemble scores 0.626 (volatility targeting 0.461, VIX rule 0.662); with 20 basis point costs, 0.622, 0.427 and 0.649. The deflated Sharpe ratio is 0.813 counting this project's 3 trials and 0.248 counting all 85. Over the registered 2020 window, from 2020-01-24 to 2020-03-24, the ensemble lost −26.8%, volatility targeting −33.0%, the VIX rule −29.2%, equal weight −36.5% and JUNIORBEES −34.5%; the best fixed action, at half exposure, lost −16.8%. JUNIORBEES set no new high in 2025 (its last peak before 2026 was on 27 September 2024), so the registered 2025 window is empty.

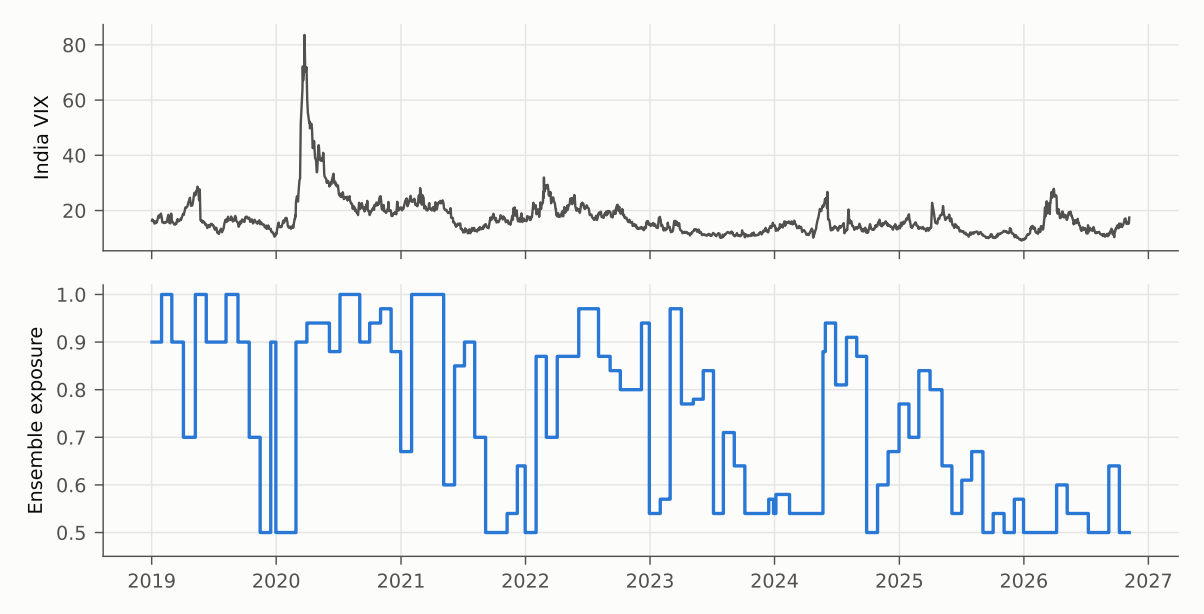


Figure 8: Study 2: India VIX (top) and the exposure chosen by the PPO ensemble (bottom).

The agent behaved as it did in Study 1 (Figure 8). Across its 97 decisions, its exposure correlates +0.42 with India VIX at the decision, against +0.62 in Study 1, and its mean exposure was 0.67 with VIX below 15 and 0.92 with VIX at 25 or above. These are descriptive reads made after the test and logged in the ledger.

8.3 The two studies side by side

Table 14: Both studies on the same test years. Intervals are 95% paired block-bootstrap intervals. The deflated Sharpe ratios use the project's final trial count for both studies.

	Study 1: Nifty 50	Study 2: Nifty Next 50
PPO − vol targeting (registered)	+0.320 [+0.000, +0.620]	+0.195 [−0.088, +0.447]
Registered verdict	supported	not supported
PPO − VIX threshold rule	+0.151 [−0.220, +0.417]	−0.028 [−0.325, +0.213]
Status of that comparison	post hoc	registered: not supported
Sharpe: PPO ensemble	0.808	0.639
Sharpe: VIX threshold rule	0.658	0.667
Sharpe: vol targeting (floor 0.5)	0.488	0.444
Sharpe: risk parity at a fixed exposure	0.611 to 0.624	0.596 to 0.602
Sharpe: equal weight	0.662	0.552
Sharpe: index ETF	0.503 (NIFTYBEES)	0.514 (JUNIORBEES)
Sharpe: random policy, median [5th, 95th pct.]	0.567 [0.410, 0.759]	0.538 [0.383, 0.688]
Max drawdown: PPO ensemble	−20.7%	−27.5%
Max drawdown: index ETF	−36.3%	−34.5%
Exposure correlation with India VIX	+0.62	+0.42
Deflated Sharpe, this project ($N = 3$)	0.916	0.813
Deflated Sharpe, with earlier work ($N = 85$)	0.408	0.248

Table 14 sets the studies side by side. The ensemble's lead over volatility targeting kept its sign but shrank from +0.320 to +0.195, and its interval now includes zero. The VIX rule, a post-hoc comparison in Study 1 and a registered one in Study 2, cannot be distinguished from the agent in either. Volatility targeting trails risk parity at every fixed exposure in both. What is consistent across the two universes is the agent's behaviour and its shallow drawdowns, not its Sharpe ratio.

9 Discussion

What the evidence supports. Two registered tests disagree. The Nifty 50 test supported H1 with no room to spare. The Nifty Next 50 test, run with every design choice frozen, did not, although the point estimate kept its sign and the ensemble led volatility targeting in every year. The VIX threshold rule, designed after the first test and registered for the second, matched the agent on both universes. We registered in advance what we would conclude from this outcome, and we conclude it. The edge over volatility targeting held on one universe only, and we find no evidence that the agent does more than a VIX threshold rule. The deflated Sharpe ratios point the same way: once our earlier work is counted, they fall below one half in both studies (Table 14). We think counting trials across a project's history, not only within its final paper, is the honest practice, and we encourage other RL studies to adopt it.

Why volatility targeting was a low bar. In both universes volatility targeting had a lower Sharpe ratio than risk parity at every fixed exposure. Its targets of 10 to 15 per cent are above the risk-parity portfolio's own volatility in calm markets, so the rule sits near full exposure and cuts exposure only after volatility has risen; in this sample that meant selling after crashes and buying back late. A policy that does anything else after a spike clears this bar easily, which is consistent with the out-of-sample weakness of volatility-managed portfolios reported by Cederburg et al. [7]. The comparisons against fixed-exposure risk parity and against the VIX rule are more informative, and the agent's advantage over both is small or absent.

What the agent learned. On both universes the agent learned contrarian volatility timing: low exposure in calm markets and high exposure after fear spikes. This is consistent with evidence that high implied volatility and variance risk premia precede higher equity returns [5, 39]. But both studies share one market path with one V-shaped crash. A policy that buys after spikes is rewarded by quick recoveries and would be punished by slow, deep declines, of which the sample has none. Its one consistent benefit, drawdowns shallower than buy-and-hold in both studies, is shared by simpler strategies: risk parity held at half exposure fell less in both.

Limitations. Both studies cover the same eight years, so Study 2 tests whether the first result depends on the particular stocks, not whether it holds in another period; the two indices correlated 0.83 before the test period, and India VIX is common to both. Six Nifty 50 members and 9 Nifty Next 50 members have no usable price series and are excluded while members. Prices come from a public vendor, checked against the exchange's daily files only for moves beyond 25 per cent. Costs are proportional, without market impact or taxes. The exposure floor of 0.5, shared by the agent, volatility targeting and the VIX rule, limits downside protection by design. We fixed the drawdown penalty at $\kappa = 1$ and did not run the A2C comparison; the ledger records these choices.

What would change the conclusion. A longer history that includes slow bear markets, such as 2008, and other markets with their own volatility indices. We would register the VIX threshold rule, or a richer rule built from the same features, as a comparator from the start. In this project the decisive comparison was one we had not thought to register first.

10 Conclusion

Moving an RL agent from return forecasting to risk control turned a clear loss into a narrow pre-registered win over volatility targeting on the Nifty 50. A second registered test, with the pipeline frozen, did not reproduce that win on the Nifty Next 50, and on both universes a two-parameter VIX threshold rule matched the agent. What the agent learned, raising exposure after fear spikes, is consistent across the two universes, but we find no evidence that the learning adds anything a simple rule does not. We report the first result as holding on one universe only. Code, data hashes, both protocols, the trial ledger and every number in this paper are in the project repository.[1]

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A Folds and per-fold selections

Table 15: Study 1 walk-forward folds. The PPO column is the mean over five seeds of the best validation Sharpe ratio. Volatility targeting, the best fixed action and the post-hoc VIX rule were each chosen on the same validation years. †Previously seen.

Test year	Train	Validate	PPO val. Sharpe	Vol tgt (target, h)	Best fixed (e, h)	VIX rule (level, h)
2019	2014–2016	2017–2018	+0.71	15%, 5	0.50, 5	20, 21
2020	2014–2017	2018–2019	−0.11	15%, 5	1.00, 5	20, 21
2021	2014–2018	2019–2020	+0.98	10%, 5	0.50, 21	25, 21
2022†	2014–2019	2020–2021	+1.64	10%, 5	0.50, 21	25, 21
2023†	2014–2020	2021–2022	+1.19	15%, 21	1.00, 21	20, 21
2024†	2014–2021	2022–2023	+1.45	15%, 5	0.50, 5	20, 21
2025†	2014–2022	2023–2024	+1.51	15%, 5	0.50, 5	15, 5
2026†	2014–2023	2024–2025	+0.58	15%, 21	1.00, 21	15, 5

Table 16: Study 2 walk-forward folds, as in the previous table; here the VIX rule is a registered baseline. Every test year was previously seen in Study 1.

Test year	Train	Validate	PPO val. Sharpe	Vol tgt (target, h)	Best fixed (e, h)	VIX rule (level, h)
2019	2014–2016	2017–2018	+1.33	15%, 5	0.50, 5	20, 21
2020	2014–2017	2018–2019	−0.11	15%, 21	0.50, 21	20, 21
2021	2014–2018	2019–2020	+0.96	10%, 21	0.50, 21	25, 21
2022	2014–2019	2020–2021	+1.82	10%, 5	0.50, 21	25, 21
2023	2014–2020	2021–2022	+0.95	15%, 21	0.50, 5	20, 21
2024	2014–2021	2022–2023	+0.92	12%, 21	0.50, 21	20, 21
2025	2014–2022	2023–2024	+1.81	10%, 21	0.50, 21	15, 5
2026	2014–2023	2024–2025	+0.77	10%, 21	1.00, 21	15, 5

B Trial ledger

Table 17: Rows of `ledger.csv`, by protocol. "Trials counted" is what enters the deflated Sharpe ratio. Development checks reran the registered configuration on fold 2019's training and validation years only and are not counted.

Row	Protocol	Kind	What	Touched test years	Trials counted
1	earlier	prior_work	Earlier PPO-MVO work	yes	76
2	earlier	prior_work	Earlier PPO-MVO work	yes	5
3	earlier	prior_work	Earlier PPO-MVO work	yes	1
4	v1	dev_smoke	Development check (fold 2019 only)	no	0
5	v1	dev_smoke	Development check (fold 2019 only)	no	0
6	v1	ppo_config	Protocol training, all folds	no	1
7	v1	test_run	The single test run	yes	0
8	v1	exploratory	Post-test analyses of the test returns	yes	0
9	v1	sensitivity	Post-hoc VIX threshold rule	yes	1
10	v1	exploratory	Descriptive reads for the manuscript	yes	0
11	v2	dev_smoke	Development check (fold 2019 only)	no	0
12	v2	ppo_config	Protocol training, all folds	no	1
13	v2	test_run	The single test run	yes	0
14	v2	exploratory	Descriptive reads for the manuscript	yes	0
15	v2	exploratory	Descriptive reads for the manuscript	yes	0

C Data repairs

Table 18: Study 1 corporate-action repairs. Demergers use the exchange's special pre-open discovered price; the split artifact is a vendor adjustment at a year boundary.

Symbol	Date	Kind	Detail
ITC	2025-01-06	demerger	discovered 455.6 vs prev close 481.6; ex-date return -4.55% → -2.84%
RELIANCE	2023-07-20	demerger	discovered 2580.0 vs prev close 2841.85; ex-date return -0.12% → +1.54%
TMPV	2025-10-14	demerger	discovered 400.0 vs prev close 660.75; ex-date return -40.15% → -1.14%
TRENT	2026-01-01	split boundary artifact	split 1.5 on 2026-06-04; raw return -33.05% → +0.43%

Table 19: Study 2: every daily move beyond 25 per cent during membership, with the vendor's return, the return from the exchange's own daily file (close over previous close) and the reviewed cause. A move counted as genuine only if both returns agreed within 1.5 percentage points and a cause was recorded. Below the line, the 4 repairs and the post-test audit of each against the exchange's file (close over the special pre-open price for the demerger).

Symbol	Date	Vendor	NSE	Reviewed cause
BANDHANBNK	2020-03-23	−25.0%	−25.0%	COVID crash: Nifty -13% on 2020-03-23
BANDHANBNK	2020-03-26	+39.3%	+39.3%	Rebound of beaten-down bank stocks at the March 2020 F&O expiry
BANKBARODA	2017-10-25	+31.4%	+31.4%	Government PSU-bank recapitalisation plan (2017-10-24)
IBULHSGFIN	2019-09-30	−34.4%	−34.4%	Delhi High Court agreed to hear a PIL alleging fund diversion; ended 34% lower
IBULHSGFIN	2020-03-19	−33.7%	−33.7%	COVID-19 crash week: housing-finance sell-off (no corporate action)
IBULHSGFIN	2020-06-19	+31.1%	+31.1%	Bulk deals, Morgan Stanley bought a 1.06% stake (2020-06-19)
IDEA	2019-07-29	−27.0%	−27.0%	Q1 FY20 results: subscriber and revenue decline
IDEA	2019-11-19	+36.0%	+36.0%	Tariff increase announced after hours on 2019-11-18
IDEA	2020-02-19	+40.0%	+40.0%	Reports that the government would not invoke bank guarantees over AGR dues
IDEA	2020-03-13	+33.3%	+33.3%	Hopes of a government relief package for telecoms
IDEA	2020-03-18	−35.0%	−35.0%	Supreme Court rejected self-assessment of AGR dues
PNB	2017-10-25	+46.2%	+46.2%	Government PSU-bank recapitalisation plan (2017-10-24)
RECLTD	2024-06-04	−25.2%	−25.2%	General-election results day: sell-off in state-owned stocks (Nifty Next 50 -9.3%)
CONCOR	2024-06-04	−19.2%	−19.2%	Split-boundary repair; repaired to +1.0%, wrong: a genuine fall; not eligible then, no effect
CONCOR	2025-01-01	−20.9%	−1.1%	Split-boundary repair; repaired to −1.1%, matches the exchange
TRENT	2026-01-01	−33.1%	+0.4%	Split-boundary repair; repaired to +0.4%, matches the exchange
VEDL	2026-04-30	−64.9%	−6.2%	Demerger: pre-open price; repaired to −6.2%, matches the exchange